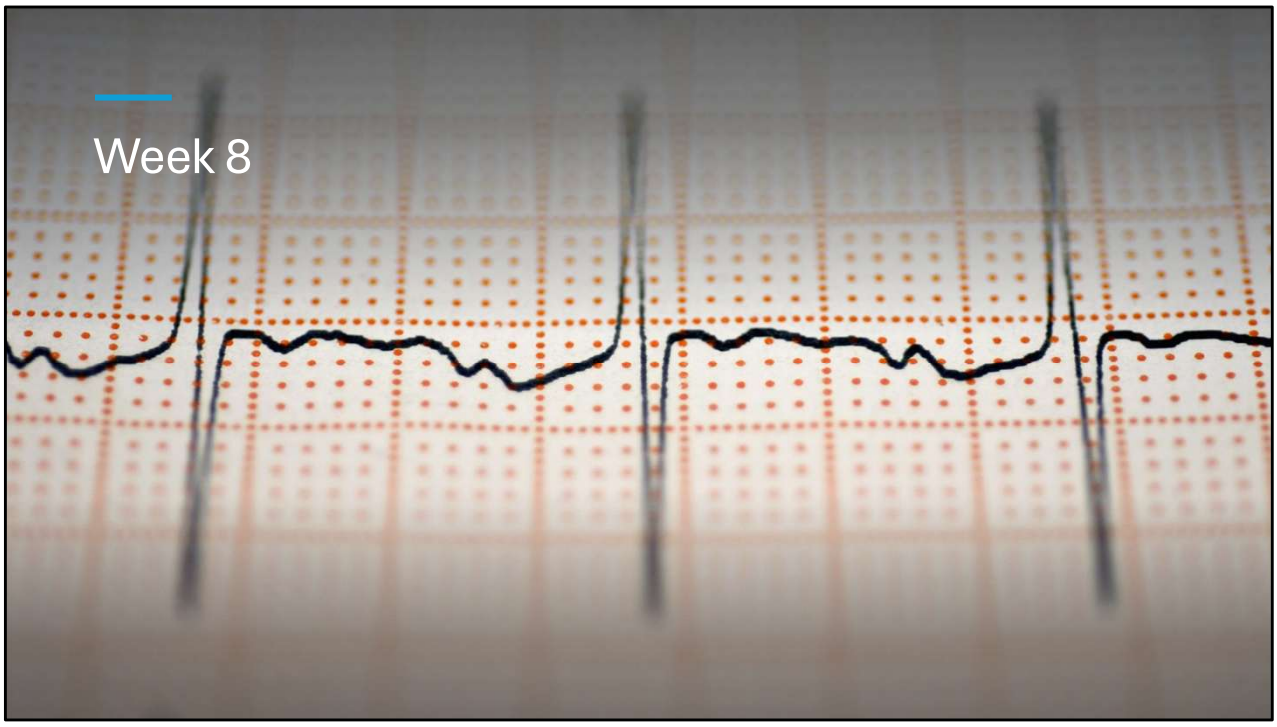
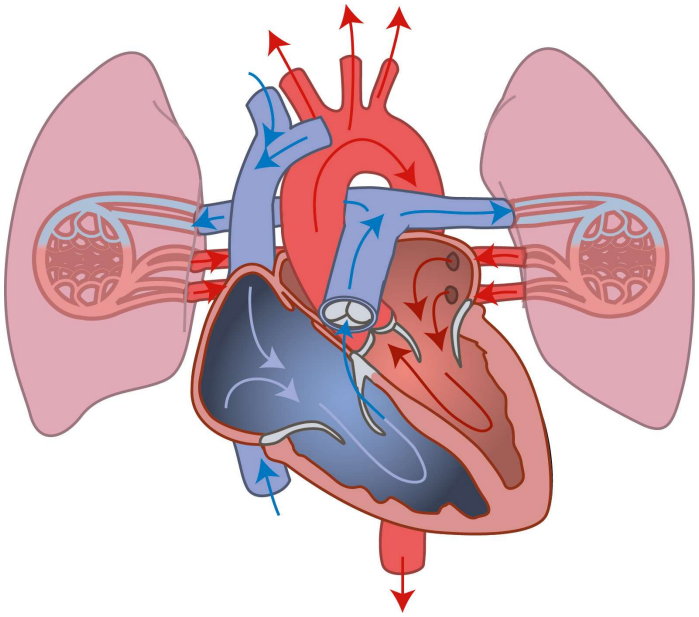
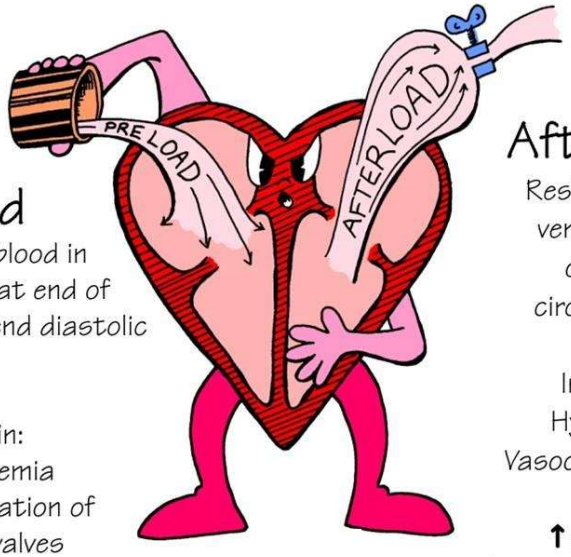

Week 8

Cardiac Review

- Blood Flow of Heart
  - HR & BP
- Preload & Afterload
  - Cardiac Output
  - Autoregulation



PRELOAD AND AFTERLOAD



Preload

Volume of blood in ventricles at end of diastole (end diastolic pressure)

Increased in:
- Hypervolemia
- Regurgitation of cardiac valves
- Heart Failure

Afterload

Resistance left ventricle must overcome to circulate blood

Increased in:
- Hypertension
- Vasoconstriction

↑ Afterload = ↑ Cardiac workload

Which of the following affect BP? Select all that apply.

A. Blood volume
B. Blood viscosity
C. Peripheral vascular resistance
D. Vasodilation
E. Force of heart contractions

All ☺

Which of the following would cause HR to increase?

A. Smoking
B. Low stress levels
C. Hypothyroidism
D. Well-trained athlete at rest

A

What is the best way to assess perfusion without advanced monitoring?

A. Skin color
B. Capillary refill
C. Blood pressure
D. Skin temperature

C

How does profound tachycardia cause decreased cardiac output?

A. The left ventricle is unable to completely fill between contractions
B. The left ventricle is unable to push against increased peripheral vascular resistance
C. There is no forceful contraction of the left ventricle in tachycardia
D. The atria and ventricles beat independently of each other

A

Name examples of the disorders that affect the components of perfusion below:

Blood Flow	Blood Volume	Oxygen	Nutrients

Blood flow – clots, polycythemia, CHF
Blood volume – dehydration, hemorrhage
Oxygen – respiratory problems, anemia
Nutrients – malnutrition, hypoglycemia

Which of the following is a complication of poor perfusion?

A. Too much oxygen delivery to tissues
B. Tissue necrosis
C. Hyponatremia
D. Weight loss

B

Hypertension

- Orthostatic Hypotension

Which of the following organs are affected by uncontrolled hypertension? Select all that apply.

A. Brain
B. Kidney
C. Heart
D. Eyes (retina)
E. Pancreas
F. Liver

A, B, C, D

Which of the following prevention measures is appropriate for a patient with a strong family history of hypertension?

A. Diet high in cholesterol
B. Regular exercise
C. Continuing to smoke 0.5 pack cigarettes daily
D. Diet high in carbohydrates

B

Which complication would be indicative of target organ damage in the presence of hypertension? Select all that apply.

A. Cataracts
B. Transient ischemic attacks (TIAs)
C. Heart failure
D. Gangrenous right foot
E. Cirrhosis

B, C, D

Which of the following are potential complication of hypertensive emergency? Select all that apply.

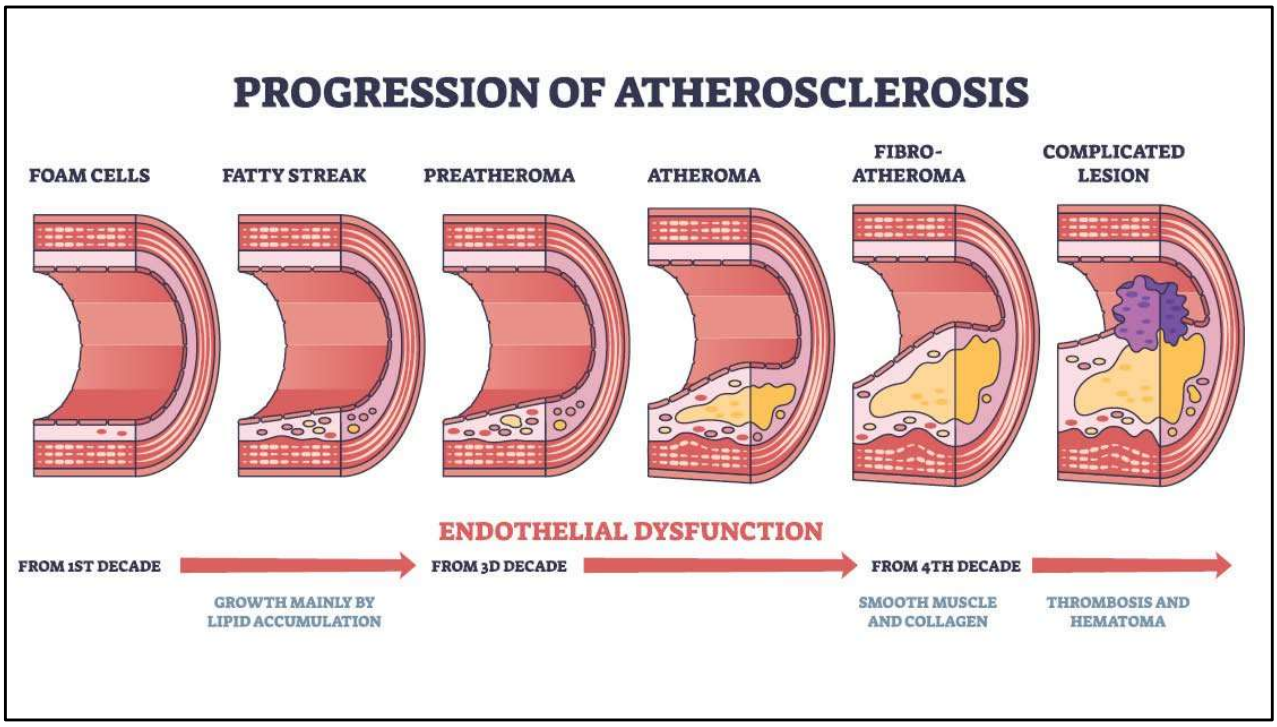
A. Rapid development of visual changes
B. Sudden onset of hemiplegia and dysarthria
C. Sharp and crushing chest pain
D. Sudden onset of right lower quadrant abdominal pain

A, B, C

Atherosclerosis

- **Peripheral Vascular Disease (PVD)**
  - Arterial Insufficiency
  - Venous Insufficiency
- **Aneurysms**
  - True vs False Aneurysms
- **Angina & Myocardial Infarction (MI)**


ATHEROSCLEROSIS
1.
Normal Artery
2.
Endothelial
Dysfunction
3.
Fatty Streak
Formation
4.
Stable (Fibrous)
Plaque Formation
5.
Unstable Plaque
Formation


PROGRESSION OF ATHEROSCLEROSIS
FOAM CELLS
FATTY STREAK
PREATHEROMA
ATHEROMA
FIBRO-ATHEROMA
COMPLICATED LESION
ENDOTHELIAL DYSFUNCTION
FROM 1ST DECADE
FROM 3D DECADE
FROM 4TH DECADE
GROWTH MAINLY BY LIPID ACCUMULATION
SMOOTH MUSCLE AND COLLAGEN
THROMBOSIS AND HEMATOMA

The term arteriosclerosis specifically refers to:

A. Intermittent vasospasm in coronary arteries
B. Development of atheromas in large arteries
C. Degeneration with loss of elasticity, narrowing, and obstruction in small arteries
D. Ischemic muscle pain in legs, particularly with exercise

C

What are modifiable risk factors that increase risk for atherosclerosis? Select all that apply.

A. Smoking
B. Sedentary lifestyle
C. Female and older than 50
D. Excluding fat and salt from diet
E. Family history of atherosclerosis
F. Hypertension and diabetes mellitus

A, B, F

A patient has a medical history of atherosclerosis. What is an atheroma?

A. A blood clot in the coronary arteries that traveled from the veins in the legs
B. A hardened accumulation of plaque, lipid, calcium, and possible clots
C. A tumor in the heart often caused by hypertension
D. A deficit in blood vessel mucosa

B

How are atherosclerosis and hypertension related in the development of arterial disease?

A. Hypertension and atherosclerosis always occur together
B. Hypertension damages arterial walls to promote development of atherosclerosis
C. High cholesterol levels influence hypertension and atherosclerosis
D. Atherosclerosis causes unstable hypertension

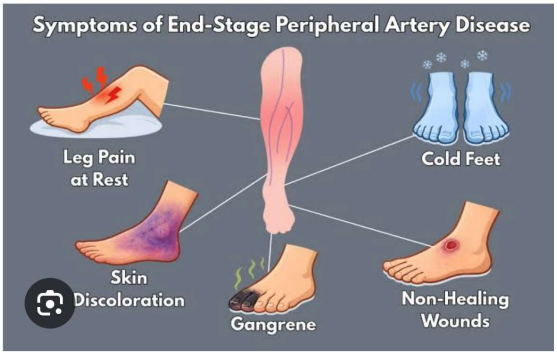
B

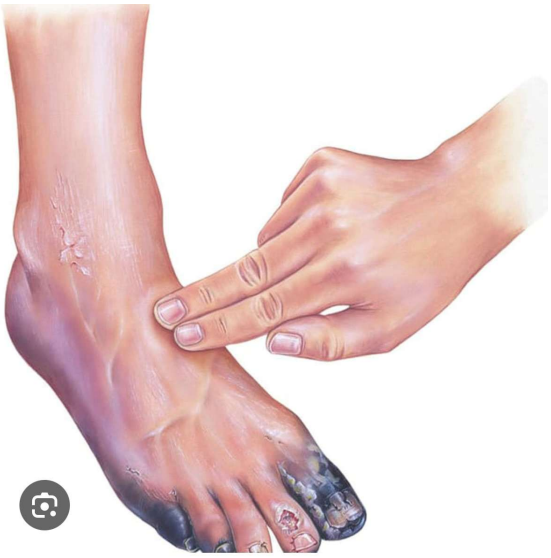
Which statements are true of plaque? Select all that apply.

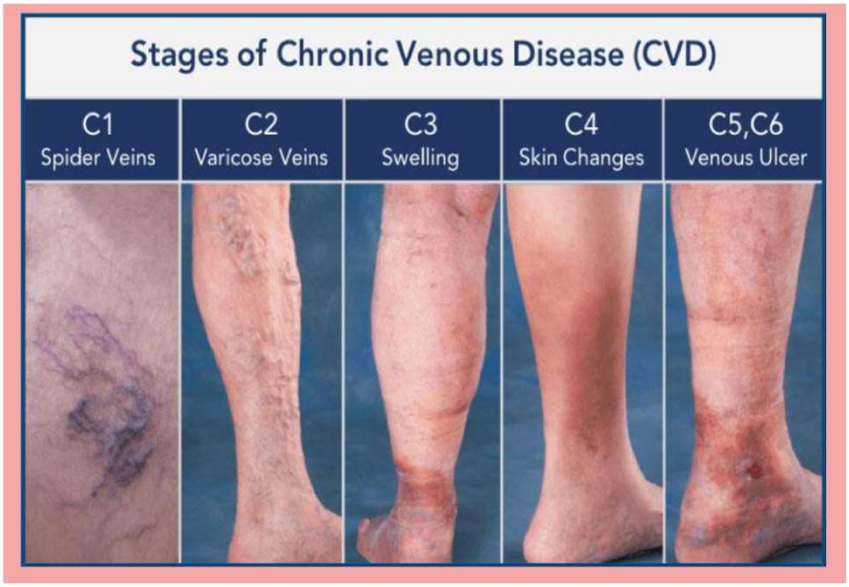
A. Stable plaques have thin caps that rupture easily
B. Stable plaques are calcified
C. Unstable plaques result in rapid thrombus formation or embolus
D. Unstable plaque is hardened and slow-growing

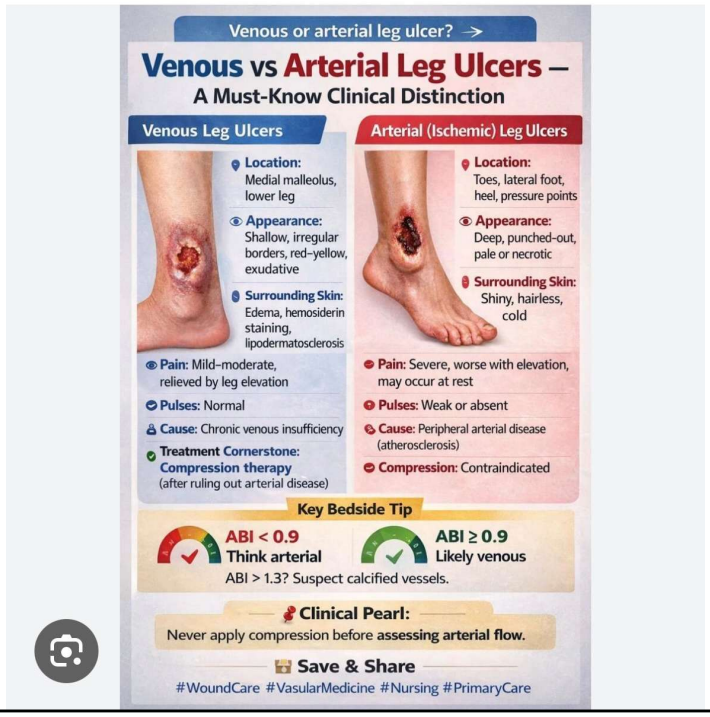
B, C

Peripheral Vascular Disease






Stages of Chronic Venous Disease (CVD)
C1
Spider Veins
C2
Varicose Veins
C3
Swelling
C4
Skin Changes
C5,C6
Venous Ulcer

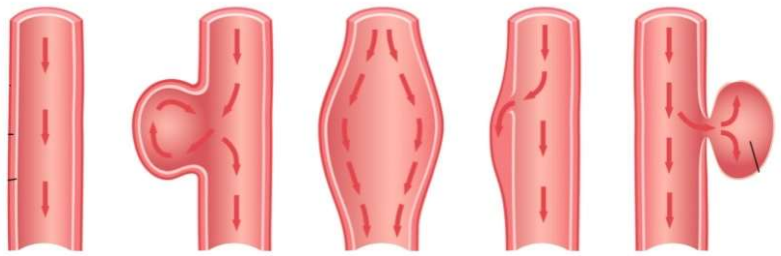

Venous or arterial leg ulcer?
Venous vs Arterial Leg Ulcers –
A Must-Know Clinical Distinction
Venous Leg Ulcers
Location: Medial malleolus, lower leg
Appearance: Shallow, irregular borders, red-yellow, exudative
Surrounding Skin: Edema, hemosiderin staining, lipodermatosclerosis
Pain: Mild–moderate, relieved by leg elevation
Pulses: Normal
Cause: Chronic venous insufficiency
Treatment Cornerstone: Compression therapy (after ruling out arterial disease)
Arterial (Ischemic) Leg Ulcers
Location: Toes, lateral foot, heel, pressure points
Appearance: Deep, punched-out, pale or necrotic
Surrounding Skin: Shiny, hairless, cold
Pain: Severe, worse with elevation, may occur at rest
Pulses: Weak or absent
Cause: Peripheral arterial disease (atherosclerosis)
Compression: Contraindicated
Key Bedside Tip
ABI < 0.9
Think arterial
ABI ≥ 0.9
Likely venous
ABI > 1.3? Suspect calcified vessels.
Clinical Pearl:
Never apply compression before assessing arterial flow.
Save & Share
#WoundCare #VasularMedicine #Nursing #PrimaryCare

Which of the following is expected in a patient with peripheral vascular disease?

A. Warm extremities
B. Orthopnea
C. Cap refill less than 3 seconds
D. Intermittent claudication

D

Aneurysms



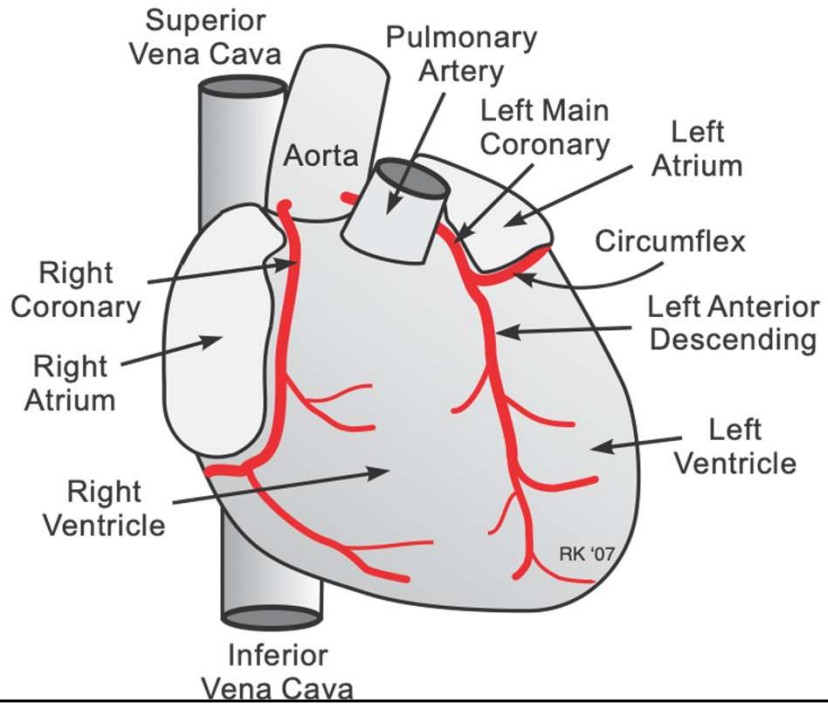
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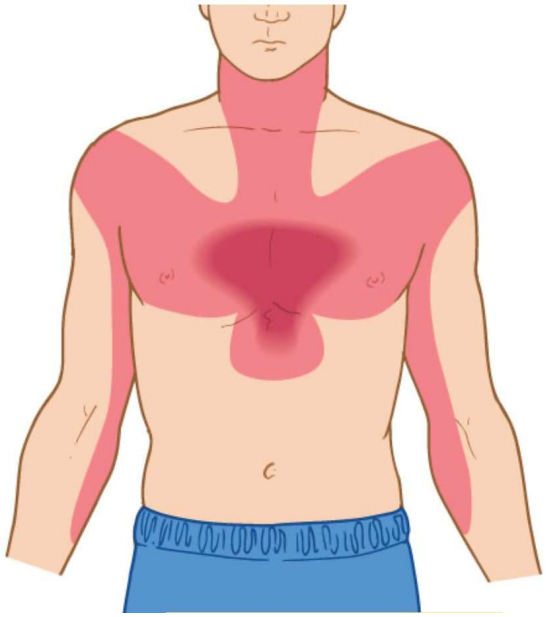
Which of the following statements is true of a dissecting aneurysm?

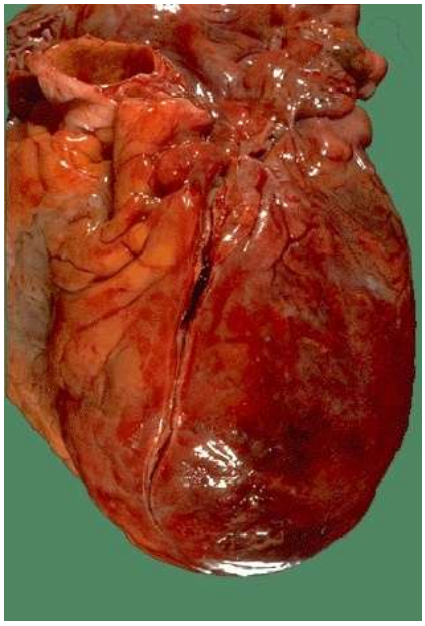
A. The blood flow within the aneurysm is completely contained in the inner vessel wall
B. The inner vessel wall becomes weak due to hypertension and separates from the outer wall, causing blood flow to escape the inner vessel
C. Congenital issues cause a weakening of the vessel wall
D. Dissecting aneurysms are most commonly found in the cerebral circulation

B

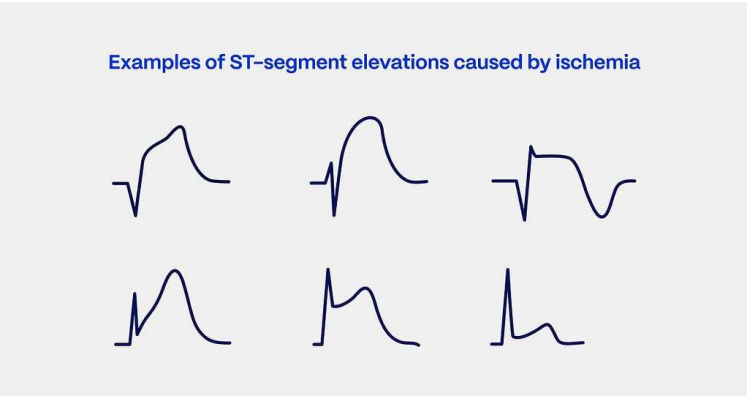
Myocardial Infarction & Angina

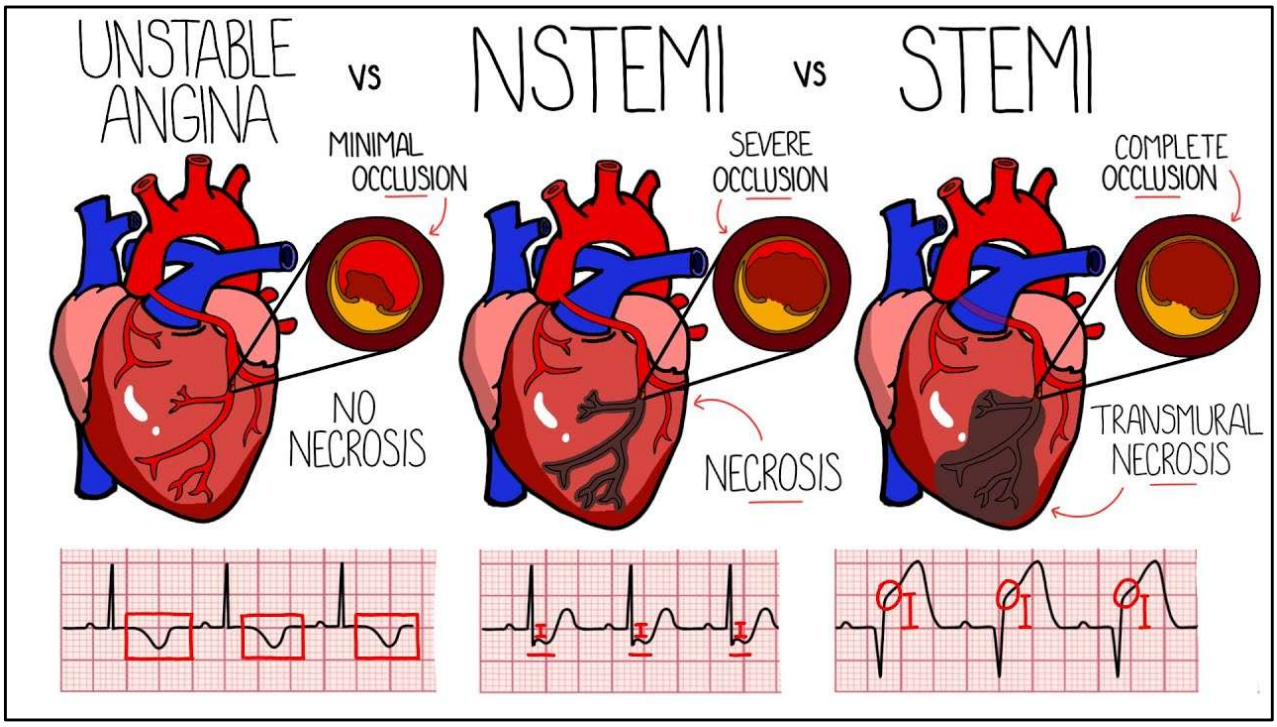






https://youtu.be/moyoYK4lZYc?si=OeEOz2EyWA2w2x25




UNSTABLE ANGINA
VS
NSTEMI
VS
STEMI
MINIMAL OCCLUSION
SEVERE OCCLUSION
COMPLETE OCCLUSION
NO NECROSIS
NECROSIS
TRANSMURAL NECROSIS

Which of the following confirms presence of a myocardial infarction?

A. A full description of the pain, including sequence of development
B. Serum isoenzymes released from necrotic cells and an EKG
C. Presence of elevated serum cholesterol and triglycerides
D. Anemia and elevated C-reactive protein

B

What is the pathophysiology of angina?

A. A deviation from normal cardiac rate or rhythm
B. Death of cardiac tissue due to ischemia
C. A deficit of oxygen to the heart muscle
D. Heart is unable to pump sufficient blood to meet the metabolic needs of the body

C

Which statement is true of angina?

A. Angina is caused by complete blockage of a coronary artery
B. Angina causes substernal chest pain that can radiate to the neck and left arm
C. A precipitating factor of angina is bradycardia
D. Angina causes necrosis of the myocardium

B

What medication is administered to patients experiencing angina?

A. Metoprolol
B. Lisinopril
C. Digoxin
D. Nitroglycerin

D

Which of the following does not cause an MI?

A. Coronary vasospasm
B. Severe hypertension
C. Anemia
D. COPD

D

Which of the following signs, symptoms, and lab values are expected in a patient with an MI?

A. Nausea
B. Hyperkalemia
C. Diaphoresis
D. Chest pain
E. Shortness of breath

All ☺

Why do embolic MIs usually cause a larger area of infarct than thrombotic MIs?

A. Thrombotic MIs occur slowly and allow for collateral circulation to develop
B. Less troponin is released with thrombotic MIs
C. Only embolic MIs are associated with irreversible myocardial damage
D. Embolic MIs are caused by atrial fibrillation

A

Which of the following statements is true regarding MIs?

A. STEMIs have ST depression on EKG
B. Only STEMIs cause elevated troponin levels
C. NSTEMIs may have T wave inversion on EKG
D. NSTEMIs are caused by prolonged and total artery occlusion

C

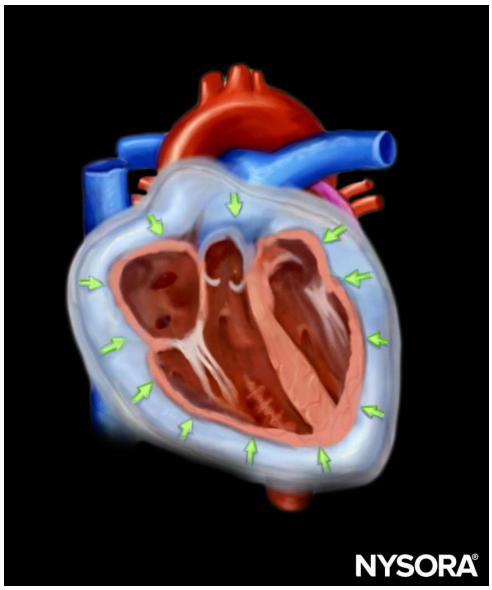
Which is not a potential complication of an acute MI?

A. Systemic vasodilation
B. Bradycardia
C. Dysrhythmias
D. Congestive heart failure (CHF)

A

Pericardial Disorders

- Pericarditis
- Pericardial Effusion/Cardiac Tamponade


NYSORA

Which of the following manifestations would you expect to find in a patient with a pericardial effusion?

A. Headache
B. Widened pulse pressure
C. Vision changes
D. Muffled heart sounds

D

Why is cardiac tamponade considered a medical emergency?

A. The patient will develop respiratory failure if untreated
B. The heart is unable to contract properly, which leads to hypotension and obstructive shock
C. The heart has increased venous return and subsequent hypervolemia
D. The patient will develop a pericardial friction rub

B

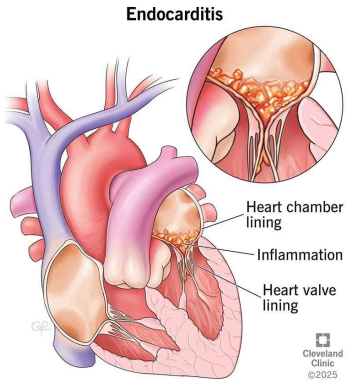
Infectious Disorders

- Infective Endocarditis
- Rheumatic Heart Disease

Endocarditis







Rheumatic Fever



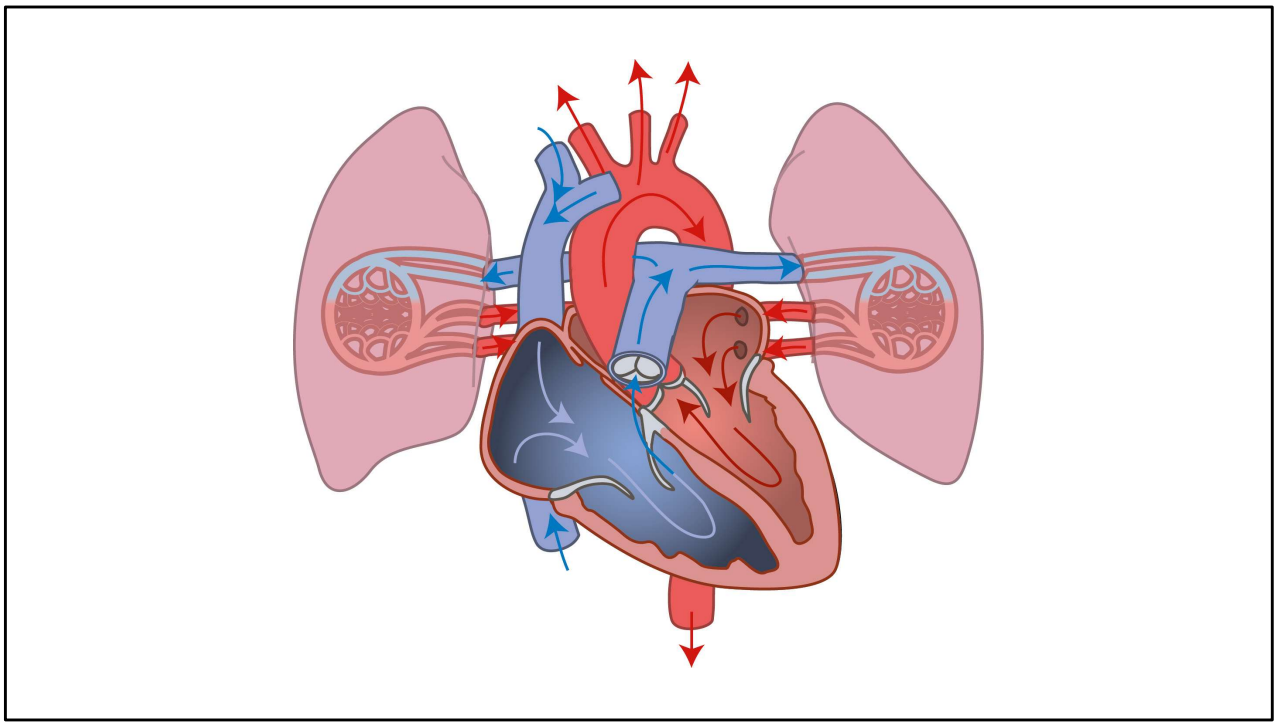
Which patient is most at risk for developing infective endocarditis?

A. A 65-year-old with a history of multiple CVAs
B. A 23-year-old IV drug user
C. A 45-year-old with recent dental procedure
D. A 50-year-old with history of diabetes mellitus

B

Valvular Disease

- Stenosis
- Regurgitation
- Prolapse (Mitral Valve)



Which of the following manifestations would you expect to see in a patient with a valvular disorder?

A. Dyspnea on exertion
B. Vision changes
C. Increased incidence of respiratory infections
D. Headache

A

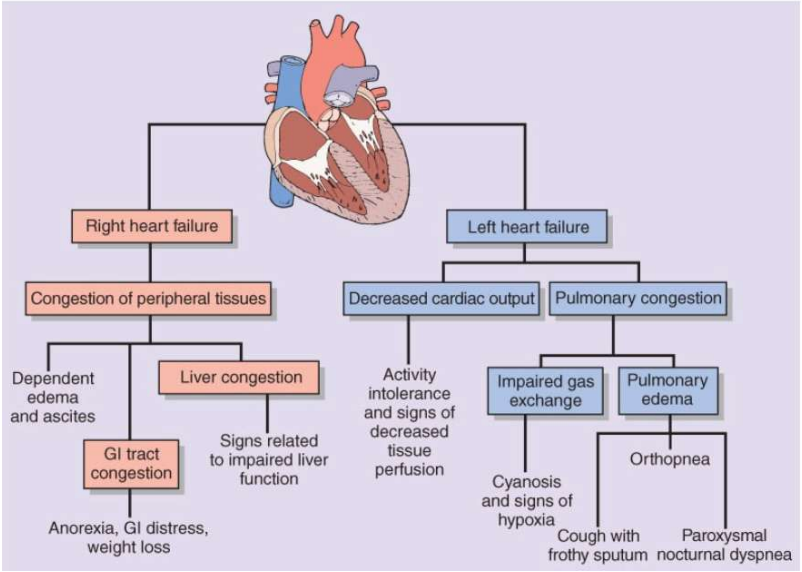
What is a complication of valve stenosis that is not a complication of valve regurgitation?

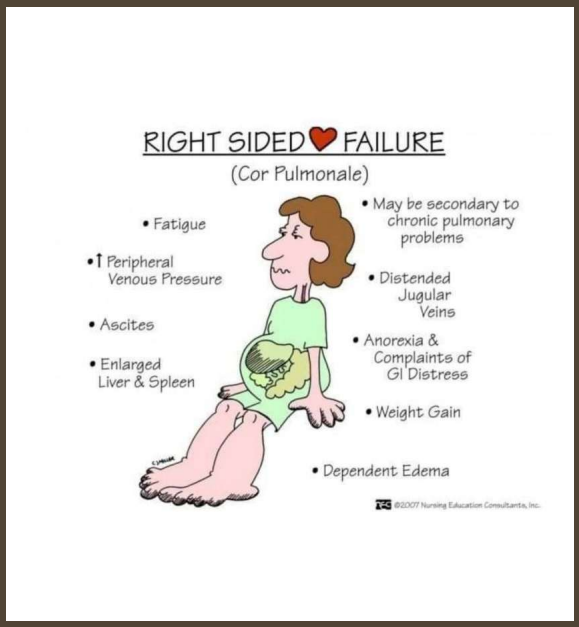
A. Heart failure
B. Pulmonary congestion
C. Orthopnea
D. Dyspnea on exertion

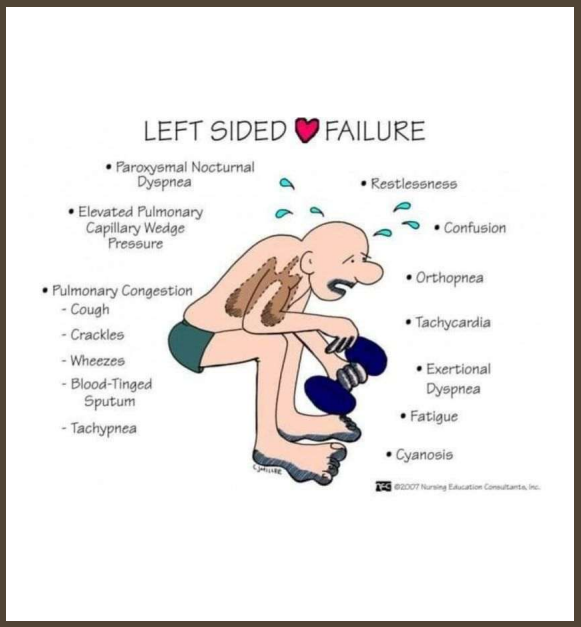
A

Heart Failure

- Diastolic vs Systolic
- Right vs Left


Right heart failure
Left heart failure
Congestion of peripheral tissues
Decreased cardiac output
Pulmonary congestion
Dependent edema and ascites
Liver congestion
GI tract congestion
Signs related to impaired liver function
Anorexia, GI distress, weight loss
Activity intolerance and signs of decreased tissue perfusion
Impaired gas exchange
Pulmonary edema
Cyanosis and signs of hypoxia
Orthopnea
Cough with frothy sputum
Paroxysmal nocturnal dyspnea


RIGHT SIDED ♥ FAILURE
(Cor Pulmonale)
• Fatigue
• ↑ Peripheral Venous Pressure
• Ascites
• Enlarged Liver & Spleen
• May be secondary to chronic pulmonary problems
• Distended Jugular Veins
• Anorexia & Complaints of GI Distress
• Weight Gain
• Dependent Edema
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LEFT SIDED ♥ FAILURE
• Paroxysmal Nocturnal Dyspnea
• Elevated Pulmonary Capillary Wedge Pressure
• Pulmonary Congestion
- Cough
- Crackles
- Wheezes
- Blood-Tinged Sputum
- Tachypnea
• Restlessness
• Confusion
• Orthopnea
• Tachycardia
• Exertional Dyspnea
• Fatigue
• Cyanosis
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The manifestations of left-sided congestive heart failure are ____, while right-sided failure has _____.

A. Edema, ascites / chest pain, dyspnea
B. Weight gain, chest pain / hepatomegaly and pulmonary edema
C. Shortness of breath and orthopnea / edema of legs and ankles
D. Intermittent leg pain / shortness of breath on exertion

C

Which of the following is considered a non-modifiable risk factor for the development of heart failure? Select all that apply.

A. Decreased age
B. Family history of cardiovascular disease
C. Congenital heart disorders
D. Cigarette smoking
E. Hyperlipidemia

B, C

Which of the following is true regarding diastolic heart failure?

A. Diastolic heart failure results in an EF <40%
B. Diastolic heart failure is more common in younger adults
C. Diastolic heart failure is a heart contraction problem
D. Diastolic heart failure often leads to systolic heart failure

D

Shock

- Cardiogenic Shock
- Hypovolemic Shock
- Distributive Shock
- Obstructive Shock

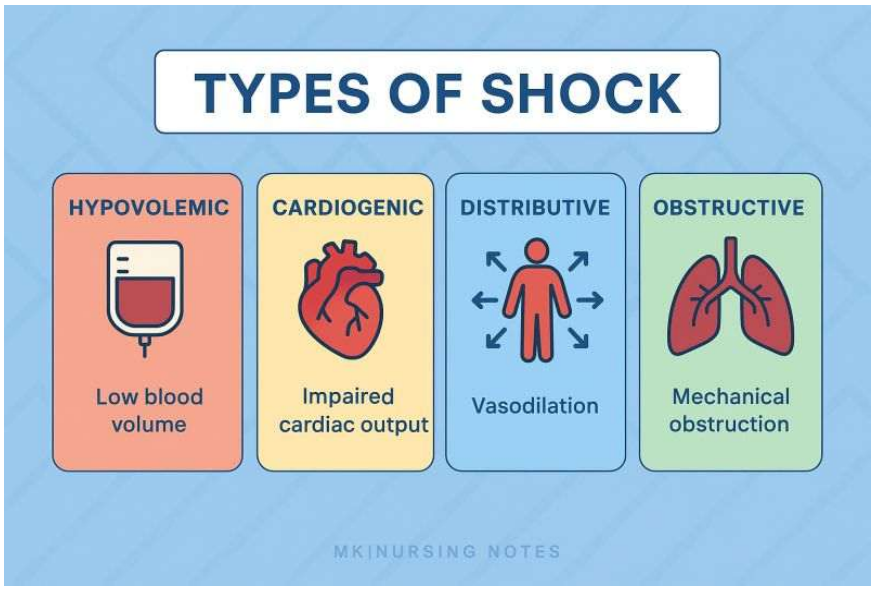

TYPES OF SHOCK
HYPOVOLEMIC
Low blood volume
CARDIOGENIC
Impaired cardiac output
DISTRIBUTIVE
Vasodilation
OBSTRUCTIVE
Mechanical obstruction
MK|NURSING NOTES

Table 1. Categories Of Shock[2]

Category	Hemodynamics	Causes
Hypovolemic	↓ preload ↑ SVR ↓ CO	Hemorrhage, GI losses, third spacing, burns
Distributive	↓ preload ↓ SVR ↑/↓ CO	Sepsis, anaphylaxis, neurogenic shock, pancreatitis
Cardiogenic	↑ preload ↑ SVR ↓ CO	Myocardial infarction, symptomatic bradycardia, valvular disease, heart blocks, end-stage heart failure
Obstructive	↓ preload ↑ SVR ↓ CO	Pulmonary embolism, tension pneumothorax, pericardial tamponade

Abbreviations: CO, cardiac output; GI, gastrointestinal; SVR, systemic vascular resistance.